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Validation Test Plan for the Digital
Time Division Command/Response
Multiplex Data Bus
Bus Controllers

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TABLE OF CONTENTS

<u>PARAGRAPH</u>	<u>TEST</u>	<u>PAGE</u>
1.	SCOPE	5
1.1	General	5
1.2	Application	5
2.	APPLICABLE DOCUMENTS	5
2.1	Standards, Military	5
3.	DEFINITIONS	5
3.1	Message	5
3.1.1	CMS - Command Message Segment	5
3.1.2	SMS - Status Message Segment	5
3.2	Bus Conditions	5
3.2.1	VSMS - Valid Status Message Segment	6
3.2.2	ISMS - Invalid Status Message Segment	6
3.2.3	NR - No Response	6
3.2.4	CS - Clear Status	6
3.3	Other Definitions	6
3.3.1	VW - Valid Word	6
3.3.2	UUT - Unit Under Test	6
3.3.3	BC - Bus Controller	6
3.3.4	RT - Remote Terminal	6
3.3.5	MCX - Mode Code	6
3.3.6	DBACC - Dynamic Bus Control Acceptance Bit	6
4.	GENERAL TEST REQUIREMENTS	6
4.1	General Monitoring Requirements	7

TABLE OF CONTENTS

(Continued)

<u>PARAGRAPH</u>	<u>TEST</u>	<u>PAGE</u>
5.	DETAILED REQUIREMENTS	7
5.1	Electrical Tests	7
5.1.1	Output Characteristics	7
5.1.1.1	Amplitude	7
5.1.1.2	Rise Time/Fall Time	8
5.1.1.3	Zero Crossing Stability	8
5.1.1.4	Distortion, Overshoot and Ringing	8
5.1.1.5	Output Symmetry	8
5.1.1.6	Output Noise	9
5.1.1.7	Output Isolation	9
5.1.1.8	Power On/Off Noise	9
5.1.2	Input Characteristics	10
5.1.2.1	Input Waveform Compatibility	10
5.1.2.1.1	Zero Crossing Distortion	10
5.1.2.1.2	Amplitude Variations	10
5.1.2.1.3	Rise and Fall Time	10
5.1.2.1.3.1	Trapezoidal	10
5.1.2.1.3.2	Sinusoidal	11
5.1.2.2	Common Mode Rejection	11
5.1.2.3	Input Impedance	11
5.2	Protocol Tests	11
5.2.1	Required Bus Controller Operation	11
5.2.1.1	Command Vocabulary Verification	12
5.2.1.2	Identification of Correct Address Field in Response	12
5.2.1.2.1	Correct Address Field with Controller-Remote Terminal Message Transfers	12
5.2.1.2.2	Correct Address Field with Remote Terminal-Remote Terminal Message Transfers	12
5.2.1.3	Minimum Response Time	13
5.2.1.3.1	Minimum Response Time: BC-RT Transfers	13
5.2.1.3.2	Minimum Response Time: RT-RT Transfers	13
5.2.1.4	Minimum No-Response Time-Out	13
5.2.1.4.1	Minimum No-Response BC-RT Transfer	13
5.2.1.4.2	Minimum No-Response Time-Out: RT-RT Transfers	13
5.2.1.5	No-Response Time-Out	14
5.2.1.5.1	No-Response Time-Out: BC-RT Transfers	14
5.2.1.5.2	No-Response Time-Out: RT-RT Transfers	14
5.2.1.6	Error Injection: BC to RT Transfers	14
5.2.1.6.1	Parity	14
5.2.1.6.1.1	Status Word Validation for Transmit Commands	15
5.2.1.6.1.2	Status Word Validation for Receive Commands	15
5.2.1.6.1.3	Data Word Validation	15
5.2.1.6.2	Word Length	15
5.2.1.6.2.1	Status Word Validation for Transmit Commands	15
5.2.1.6.2.2	Status Word Validation for Receive Commands	16

TABLE OF CONTENTS

(Continued)

<u>PARAGRAPH</u>	<u>TEST</u>	<u>PAGE</u>
5.2.1.6.2.3	Data Word Validation	16
5.2.1.6.3	Bi-Phase Encoding	16
5.2.1.6.3.1	Status Word Validation for Transmit Commands	16
5.2.1.6.3.2	Status Word Validation for Receive Commands	17
5.2.1.6.3.3	Data Word Validation	17
5.2.1.6.4	Sync Encoding	17
5.2.1.6.4.1	Status Word Sync Validation	17
5.2.1.6.4.2	Data Word Sync Validation	18
5.2.1.6.5	Response Message Length	18
5.2.1.6.5.1	Response to Receive Commands	18
5.2.1.6.5.2	Response to Transmit Commands	18
5.2.1.6.6	Discontiguous Data	19
5.2.1.7	Error Injection: RT to RT Transfers	19
5.2.1.7.1	Parity	19
5.2.1.7.1.1	Status Word Validation for Responses to RT to RT Transmit Commands	20
5.2.1.7.1.2	Status Word Validation for Responses to RT to RT Receive Commands	20
5.2.1.7.1.3	RT to RT Data Word Validation	20
5.2.1.7.2	Word Length	20
5.2.1.7.2.1	Status Word Validation for Responses to RT to RT Transmit Commands	21
5.2.1.7.2.2	Status Word Validation for Responses to RT to RT Receive Commands	21
5.2.1.7.2.3	Data Word Validation	21
5.2.1.7.3	Bi-Phase Encoding	22
5.2.1.7.3.1	Status Word Validation for Responses to RT to RT Transmit Commands	22
5.2.1.7.3.2	Status Word Validation for Responses to RT to RT Receive Commands	22
5.2.1.7.3.3	Data Word Validation	22
5.2.1.7.4	Sync Encoding	23
5.2.1.7.4.1	Status Word Sync Validation for Response to RT to RT Transmit Commands	23
5.2.1.7.4.2	Status Word Sync Validation for Response to RT to RT Receive Commands	23
5.2.1.7.4.3	Data Word Sync Validation	23
5.2.1.7.5	Response Message Length	24
5.2.1.7.5.1	Message Length Validation for RT to RT Transmit Commands	24
5.2.1.7.5.2	Message Length Validation for RT to RT Receive Commands	24
5.2.1.7.6	Discontiguous Data	25
5.2.2	Optional Operation	25
5.2.2.1	Mode Commands	25
5.2.2.1.1	Dynamic Bus Control	25
5.2.2.1.2	Mode Commands Without Data Words	26
5.2.2.1.3	Transmit Mode Commands With Data Words	27



TABLE OF CONTENTS

(Continued)

<u>PARAGRAPH</u>	<u>TEST</u>	<u>PAGE</u>
5.2.2.1.4	Receive Mode Commands With Data Words	28
5.2.2.2	Status Word	28
5.2.2.2.1	Status Bits, Receive Commands	28
5.2.2.2.2	Status Bits, Transmit Commands	29
5.2.2.2.3	Status Bits, RT to RT Commands, Transmitter Status	29
5.2.2.2.4	Status Bits, RT to RT Commands, Receiver Status	29
5.2.2.2.5	Status Bits, Mode Command	29
5.2.2.3	Broadcast Command	29
5.2.2.3.1	Broadcast Receive Commands	29
5.2.2.3.2	Broadcast Receive Command With Response	30
5.2.2.3.3	Broadcast RT-RT Command	30
5.2.2.3.3.1	Broadcast RT-RT Command With No-Response	30
5.2.2.3.3.2	Broadcast RT-RT Command With Response	30
5.2.2.3.3.3	Normal Broadcast RT-RT Command	30
5.2.2.3.4	Broadcast Mode Commands	30
5.2.2.3.5	Broadcast Mode Command With Response	30
5.2.2.4	Response On Alternate Bus	31
5.3	Noise Rejection	31

LIST OF TABLES

<u>TABLE</u>	<u>TITLE</u>	<u>PAGE</u>
1	Mode Codes	32
2	Criteria for Acceptance or Rejection of a Bus Controller for the Noise Rejection Test	33

LIST OF FIGURES

<u>FIGURE</u>	<u>TITLE</u>	<u>PAGE</u>
1	General Test Configuration	34
2	Waveform Measurements	35
3	Zero Crossing Interval Measurements	36
4	Output Noise Configuration	37
5	Output Isolation	38
6A	Transformer Coupled Common Mode Configuration	39
6B	Direct Coupled Common Mode Configuration	40
7	Gap Time Measurement	41
8	Configuration for Noise Rejection Test	42

LIST OF APPENDICES

<u>APPENDIX</u>	<u>TITLE</u>	<u>PAGE</u>
A	Test Plan Changes for MIL-STD-1553B, Notice 2 BCs	43

1. SCOPE:

This test plan is broken into three major sections for the testing of bus controllers Electrical, Protocol and Noise tests.

- 1.1 General: This validation test plan defines the test requirements for verifying that the design of bus controllers meets the requirements of MIL-STD-1553B, "Digital Time Division Command/Response Multiplex Data Bus."
- 1.2 Application: This is a general test plan for any first article, prototype, or engineering preproduction bus controller designed to meet the requirements of MIL-STD-1553B. These requirements shall apply to the bus controller under test, when invoked in a specification or statement of work.

This document deals only with bus controller functions that are explicitly described in MIL-STD-1553B. This document does not specify tests that check the integrity of data from the subsystem to the bus, nor does it attempt to address the system test requirements.

2. APPLICABLE DOCUMENTS:

2.1 Standards. Military:

MIL-STD-1553B, 21 Sep 1978 "Aircraft Internal Time Division Command/Response Multiplex Data Bus" with Notice 2, 8 September 1986.

3. DEFINITIONS:

- 3.1 Message: A single message is the transmission of a command word, status word, and data words if they are specified. For the case of a remote terminal to remote terminal (RT to RT) transmission, the message shall include the two command words, the two status words, and data words. All messages are divided into two message segments: a command message segment and a status message segment.
 - 3.1.1 CMS - Command Message Segment: A command message segment consists of a command word and associated data words, if any. In the case of RT to RT transfer, the command message segment includes both the receive command word and the transmit command word.
 - 3.1.2 SMS - Status Message Segment: A status message segment consists of a status word and associated data words, if any. In the case of RT to RT transfer, the status message segment includes the status word of the transmitting RT, its associated data words, and the status word of the receiving RT.
- 3.2 Bus Conditions: The following are definitions of bus conditions that bus controllers are required to detect. A description of the actual signal, state, or action taken within the bus controller, which is used during test procedures to indicate the detection of each of the following bus conditions, shall be attached to the test results. Bus controller states may or may not be distinguishable; that is, a single bus controller signal may indicate both NR and ISMS.

- 3.2.1 VSMS - Valid Status Message Segment: A VSMS condition occurs within a bus controller when it determines that a SMS contains only a CS and valid data words, the data words are contiguous to the status word and each other, and the response time is as specified in MIL-STD-1553.
- 3.2.2 ISMS - Invalid Status Message Segment: An ISMS condition occurs within a bus controller when it determines that a SMS is not a VSMS.
- 3.2.3 NR - No Response: A no response condition occurs if a bus controller determines that there has been no response to a command.
- 3.2.4 CS -Clear Status: The status word may have the busy bit or service request bit set, or both. The status word must have the correct terminal address, all other status code bits in the status word must be zero and the associated message segment must have the proper word count.

3.3 Other Definitions:

- 3.3.1 VW - Valid Word: A valid word conforms to the following minimum criteria:

- a. The word begins with a valid sync field.
- b. The bits are in a valid Manchester II code.
- c. The information field has 16 bits plus parity.
- d. The word parity is odd.

- 3.3.2 UUT - Unit Under Test: For the purposes of this test plan, UUT refers to the bus controller under test.

- 3.3.3 BC - Bus Controller:

- 3.3.4 RT - Remote Terminal:

- 3.3.5 MCX - Mode Code: For the purposes of this test plan, mode codes are tabulated in Table 1.

- 3.3.6 DBACC - Dynamic Bus Control Acceptance Bit: The status word bit at bit time 18.

4. GENERAL TEST REQUIREMENTS:

The following paragraphs define the configurations, pass/fail criteria, and general procedures for testing bus controllers. Specifically, this document contains the test configurations and procedures for the Electrical Tests (5.1), the Protocol Tests (5.2), and the Noise Rejection Test (5.3) for MIL-STD-1553 bus controllers. Bus controllers that can be deployed in more than one configuration, depending upon the choice of stored program or hardware option, shall be tested using all specified configurations such that the test program exercises all of the bus controller's capabilities. Before performing each test, the UUT shall be cleared of any preexisting error conditions to insure that all test results are independent of previous results.

4.1 General Monitoring Requirements: In addition to the specific tests that follow, certain parameters of the BC transmission must be continuously monitored throughout all tests. These parameters are:

1. contiguous data
2. proper Manchester encoding
3. proper bit count
4. odd parity
5. proper word count
6. proper intermessage gap times
7. proper sync
8. proper CMS transmission

This general monitoring requirement shall include the verification that the CMS transmitted by the UUT is the CMS specified to be transmitted by the procedure.

The UUT shall have failed the test if, at any time during the test, any of these parameters fail to meet the requirements of MIL-STD-1553. Record the parameters for all failures.

5. DETAILED REQUIREMENTS:

5.1 Electrical Tests: All tests shall use Fig. 1, General Test Configuration, with all measurements taken at point "A", unless otherwise noted. Each test paragraph contains the requirements for both transformer and direct coupled stubs. A UUT which provides both transformer and direct coupled stubs must be tested on both stubs. Electrical tests shall be performed on all buses for UUTs with redundant bus configurations.

5.1.1 Output Characteristics: The following tests are designed to verify that all UUT output characteristics comply with MIL-STD-1553.

5.1.1.1 Amplitude: A receive command shall be transmitted by the UUT, with the maximum number of words that it is capable of sending. The amplitude of the waveform transmitted by the UUT shall be measured, peak-to-peak, as shown in Fig. 2.

The pass criterion for V_{pp} for transformer coupled stubs is 18.0 V minimum and 27.0 V maximum.

The pass criteria for V_{pp} for direct coupled stubs are 6.0 V minimum and 9.0 V maximum.

- 5.1.1.2 Rise Time/Fall Time: A receive command shall be sent by the UUT with at least one data word. The rise and fall time of the UUT waveform shall be measured between the 10 and 90% points of the waveform as shown in Fig. 2. The measurements shall be taken at both the rising and falling edges of a sync waveform and a data bit waveform. The rise time (T_R) and the fall time (T_F) shall be measured.

The pass criteria are $100 \text{ ns} \leq T_R \leq 300 \text{ ns}$ and $100 \text{ ns} \leq T_F \leq 300 \text{ ns}$.

The measured parameters, T_R and T_F , shall be recorded.

Note: The rise time of the sync waveform shall be measured at the mid-crossing of a data word sync, and the fall time of the sync waveform shall be measured at the mid-crossing of the command word sync. The rise and fall times of the data bit waveform shall be measured at a zero crossing where the prior zero crossing and next zero crossing are at 500 ns intervals from the measured zero crossing.

- 5.1.1.3 Zero Crossing Stability: A receive command shall be sent by the UUT, with at least one data word having zero crossing time intervals of 500, 1000, 1500, and 2000 ns. The zero crossing time (T_{ZC}) shall be measured for both the positive to negative and the negative to positive waveforms as shown in Fig. 3.

The pass criteria are that $T_{ZC} = 500 + 25 \text{ ns}$, $1000 + 25 \text{ ns}$, $1500 + 25 \text{ ns}$ and $2000 + 25 \text{ ns}$.

The measured parameters, T_{ZC} , shall be recorded for each case.

- 5.1.1.4 Distortion, Overshoot and Ringing: A receive command shall be sent by the UUT with at least 1 data word. The distortion of the waveform, distortion voltage (V_D), shall be measured as indicated in Fig. 2.

The pass criteria are $V_D \leq 900 \text{ mV}$, for transformer coupled stubs or $V_D \leq 300 \text{ mV}$, for direct coupled stubs.

The worst measured parameter, V_D , shall be recorded.

- 5.1.1.5 Output Symmetry: A receive command shall be sent by the UUT with the maximum number of data words that the UUT is capable of transmitting. The output symmetry is determined by measuring the waveform tail-off at the end of each message. The maximum residual voltage (V_R) shall be measured as shown in Fig. 2. The minimum intermessage gap time shall be 1 Ms.

The pass criteria are $V_R \leq 250 \text{ W}$ peak for transformer coupled stubs and $V_R \leq 90 \text{ W}$ peak for direct coupled stubs after time T_T (the time beginning 2.5 μs after the midbit zero crossing of the last parity bit)

5.1.1.5 (Continued):

This test shall be run six times with each data word in the message having the same bit pattern. The six data word bit patterns that shall be used are:

8000(HEX), 7FFF(HEX), 0000(HEX), UFF(HEX), 5555(HEX), and AAAA(HEX)

The measured parameter, VR, shall be recorded for each bit pattern.

5.1.1.6 Output Noise: The test configuration shown in Fig. 4 shall be used to test the UUT inactive bus output noise levels. The test shall be conducted while the UUT is in the power-on receive state and the power-off state. The output noise (Vrms) shall be measured at point "A" as shown in Fig. 4 for both states. Measurements shall be made with an instrument that has a minimum frequency bandwidth of from DC to 10 MHz.

The pass criteria are Vrms $\leq$ 14.0 mV for transformer coupled stubs and Vrms $\leq$ 5.0 mV for direct coupled stubs.

The measured parameter, Vrms, shall be recorded for each case.

5.1.1.7 Output Isolation: This test shall be performed only if the UUT is configured with redundant buses. A receive command shall be sent by the UUT with the maximum number of data words that it is capable of sending. The voltage of the output waveform transmitted by the UUT shall be measured on the active and redundant bus (or buses). Each data bus shall be alternately activated and measurements taken.

The pass criterion is that the ratio in dB between the output voltage on the active bus and the output voltage on all inactive buses shall be greater than or equal to 45 dB (Fig. 5).

The measured parameter, output isolation, expressed as a ratio in dB, shall be recorded for each bus combination.

5.1.1.8 Power On/Off Noise: A UUT shall limit any spurious differential output during a power-up or power-down sequence. Power shall be applied to the UUT and any outputs from the UUT onto the bus shall be measured. Power shall be removed from the UUT and any output onto the bus from the UUT shall be measured. Repeat the test ten times.

The pass-criteria are:

- a. For transformer coupled stubs, any spurious noise pulses produced shall be less than +250 mV peak.
- b. For direct coupled stubs, any spurious noise pulses produced shall be less than +90 W peak.

All measured parameters, output noise amplitudes and pulse widths, shall be recorded.

Note: This test shall be performed using the normal on/off power sequence to the UUT.

5.1.2 Input Characteristics: The input tests are designed to verify that multiplex devices can properly decode bi-phase data.

5.1.2.1 Input Waveform Compatibility:

5.1.2.1.1 Zero Crossing Distortion: A transmit message shall be transmitted by the UUT and the test equipment shall respond appropriately. Positive and negative zero crossing distortions equal to 150 ns minimum, with respect to the previous zero crossing, shall then be introduced individually to each zero crossing of each word transmitted to the UUT in response to the transmit command.

The transmitted signal amplitude at point "A", Fig. 1, shall be 2.1 Vpp for transformer coupled stubs and 3.0 Vpp for direct coupled stubs. Each zero crossing distortion shall be transmitted to the UUT a minimum of 1000 times. There shall be only one zero crossing deviation per message.

The pass criterion is the detection of a VSMS by the UUT for each SMS with a zero crossing distortion.

Positive and negative zero crossing distortions shall then be applied in turn to a single zero crossing and adjusted to determine the values at which the first ISMS is detected by the UUT. Record these values.

5.1.2.1.2 Amplitude Variations: A transmit command requesting the maximum number of data words specified by the UUT shall be transmitted by the UUT to the test equipment. The voltage of the test equipment response shall be decremented from 6.0 Vpp to 0.1 Vpp for transformer coupled stubs and from 9.0 Vpp to 0.1 Vpp for direct coupled stubs in steps no greater than 0.1 Vpp. A minimum of 1000 words shall be transmitted for each setting.

The pass criteria are:

- a. A VSMS for each SMS with $0.86 \leq V_{pp} \leq 6.0$ for transformer coupled stubs and $1.2 \leq V_{pp} \leq 9.0$ for direct coupled stubs
- b. A NR for each SMS with $V_{pp} \leq 0.20$ for transformer coupled stubs and $V_{pp} \leq 0.28$ for direct coupled stubs

The measured parameter, Vpp, at which NR first occurs shall be recorded.

5.1.2.1.3 Rise and Fall Time:

5.1.2.1.3.1 Trapezoidal: A transmit message involving one or more data words shall be sent by the UUT. The test equipment shall respond with a signal amplitude of 2.1 Vpp for the transformer coupled stub and 3.0 Vpp for the direct coupled stub. The rise and fall times shall be less than or equal to 100 ns, as shown in Fig. 2.

The pass criterion is a VSMS by the UUT for each SMS transmitted.

- 5.1.2.1.3.2 Sinusoidal: A transmit message involving one or more data words shall be sent by the UUT. The test equipment shall respond with a signal amplitude of 2.1 Vpp for the transformer coupled stub and 3.0 Vpp for the direct coupled stub. The rise and fall times of the signal shall approximate that of a 1 MHz sinusoidal signal.

The pass criterion is a VSMS by the UUT for each SMS transmitted.

- 5.1.2.2 Common Mode Rejection: The Common Mode Test Configuration, Fig. 6A and Fig. 6B, shall be used for this test. Transmit commands with the UUT's maximum word count shall be transmitted by the UUT at the maximum rate consistent with the minimum intermessage gap defined in 4.3.3.7 of MIL-STD-1553. The test equipment shall respond with a common mode voltage injected at point "C", and the UUT response observed. The voltage of the transmitted message measured at point "A" shall be 0.86 Vpp for transformer coupled stubs and 1.2 Vpp for direct coupled stubs. The following common mode voltage levels shall be applied in turn; +10.0 V.D.C. to ground, -10.0 V.D.C. to ground and a ± 10 Vp line to ground sinusoidal signal that is swept through the range of 1 Hz to 2 MHz. Each test condition shall be present for a minimum time period of 90 S.

The pass criterion is a VSMS by the UUT for each SMS transmitted at each setting.

If a failure occurs, the frequency of the common mode signal injected shall be recorded.

- 5.1.2.3 Input Impedance: The input impedance of the UUT shall be measured with the UUT power on and with the UUT power off. The input impedance, Z_{in} , shall be measured with a sinusoidal waveform having an amplitude 1.0 Vrms to 2.0 Vrms, at the following frequencies: 75.0 kHz, 100.0 kHz, 250.0 kHz, 500.0 kHz and 1.0 MHz.

The pass criteria are $Z_{in} \geq 1000 \Omega$ for transformer coupled stubs and $Z_{in} \geq 2000 \Omega$ for direct coupled stubs.

The measured parameter, Z_{in} , shall be recorded at each frequency.

- 5.2 Protocol Tests: All tests in this section shall use the test configuration shown in Fig. 1. The test signal amplitude shall be $3.0 \text{ Vpp} \pm 0.1 \text{ Vpp}$ for direct coupled stubs and $2.1 \text{ Vpp} \pm 0.1 \text{ Vpp}$ for transformer coupled stubs measured at point A. For UUTs having both direct and transformer coupled stubs, the protocol tests need only be performed on one stub type per bus. Any condition which causes the UUT to respond other than as called out in MIL-STD-1553, to lock up or require a power cycle in order to recover from a failure, shall automatically cause that UUT to fail the test. The protocol tests shall be performed on all buses for UUTs with redundant bus configurations.

- 5.2.1 Required Bus Controller Operation: The following tests verify all required operations of a BC:

5.2.1.1 Command Vocabulary Verification: This test verifies that the UUT is capable of transmitting every command word required by its design. The UUT shall transmit each unique command word that it is designed to transmit. Receive command words shall be followed by the correct number of data words. The pass criterion is the verification of the UUT's transmission of all command words required by its design.

5.2.1.2 Identification of Correct Address Field in Response: This test verifies that the UUT accepts valid message responses with the correct address field and identifies illegal message responses. Remote terminal address field "31" (11111) shall not be transmitted by the UUT for this test.

5.2.1.2.1 Correct Address Field with Controller-Remote Terminal Message Transfers: This test verifies that the UUT detects an error when the address field in a SMS does not match the address in its associated CMS. The UUT shall send 32 receive commands and 32 transmit commands all with the same RT address to the test equipment. The test equipment shall respond to each CMS with an SMS. Each SMS will have a status word with a different RT address so that each one of the 32 possible addresses is used once in response to a receive command, and once in response to a transmit command. The above procedure shall be performed for each implemented RT address in the command word, so that each RT address (except "11111") is used once. This test requires that the UUT send a total of $N*64$ CMSs where N is the number of RT addresses implemented by the UUT.

The pass criteria are an ISMS for each SMS with an incorrect address in the status word and a VSMS for each SMS with the correct address in the status word.

5.2.1.2.2 Correct Address Field with Remote Terminal-Remote Terminal Message Transfers: This test verifies that the UUT detects an error when the address field in a SMS does not match the address in its associated CMS for RT-RT message transfers. The UUT shall send $(31*31)-31=930$ unique, RT-RT command pairs so that every possible combination of transmitting and receiving addresses is used. The test equipment shall respond to each command pair with the correct address in the transmitting RT's status word, and each possible address in the receiving RT's status word. The above procedure shall be performed so that each implemented incorrect address and the single correct address are used in the receiving RT's status word. The above test shall then be repeated using the correct address in the receiving RT's status word, and each implemented address in each transmitting RT's status word. This test requires that the UUT shall send a total of $930*N*2$ command pairs, where N is the number of RT addresses implemented by the UUT.

The pass criteria are an ISMS for each SMS with an incorrect address in a status word and a VSMS for each SMS with correct status word addresses.

5.2.1.3 Minimum Response Time: This test verifies that the UUT validates responses with minimum response times.

5.2.1.3.1 Minimum Response Time: BC-RT Transfers: The UUT shall send a transmit command. The test equipment shall respond with a valid status and the number of valid data words requested at time T equal to 4.0 μ s maximum after the transmit command as shown in Fig. 7. A minimum of 1000 messages shall be sent during this test. The UUT shall then send a valid receive command with the correct number of valid data words. The test equipment shall respond with a valid status word starting with time T equal to 4.0 μ s after the last data word as shown in Fig. 7. A minimum of 1000 messages shall be sent during this test.

The pass criterion is a VSMS for each SMS transmitted.

5.2.1.3.2 Minimum Response Time: RT-RT Transfers: The UUT shall send an RT-RT CMS. The test equipment shall respond with a status word and the number of data words requested at time T equal to 4.0 μ s maximum after the receive command as shown in Fig. 7. The test equipment shall send a second status word for the receiving terminal at time T equal to 4.0 μ s maximum after the last data word associated with the transmitting RT's response as shown in Fig. 7. A minimum of 1000 messages shall be sent during this test.

The pass criterion is a VSMS after each SMS transmitted.

5.2.1.4 Minimum No-Response Time-Out: This test verifies that the UUT does not indicate a no-response before the minimum no-response time.

5.2.1.4.1 Minimum No-Response BC-RT Transfer: The UUT shall send a transmit command. The test equipment shall respond with a valid status and the number of valid data words requested, at time T equal to 14.0 μ s minimum after the transmit command as shown in Fig. 7. A minimum of 1000 messages shall be sent during this test. The UUT shall then send a valid receive command with the correct number of valid data words. The test equipment shall respond with a valid status word at time T equal to 14.0 μ s minimum after the last data word. The UUT shall detect correct test equipment response. A minimum of 1000 messages shall be sent during this test.

The pass criterion is a VSMS after each SMS transmitted.

5.2.1.4.2 Minimum No-Response Time-Out: RT-RT Transfers: The UUT shall send an RT-RT CMS. The test equipment shall respond with a status word and the number of data words requested at time T equal to 14.0 μ s minimum after the transmit command as shown in Fig. 7. The test equipment shall send a second status word for the receiving terminal at time T equal to 14.0 μ s minimum after the last data word associated with the transmitting RT's response, as shown in Fig. 7. A minimum of 1000 messages shall be sent during this test.

The pass criterion is a VSMS after each SMS transmitted.

5.2.1.5 No-Response Time-Out: This test verifies that the UUT detects a no-response when an RT does not respond.

5.2.1.5.1 No-Response Time-Out: BC-RT Transfers: The UUT shall send a minimum of six command messages, three with receive commands and data and three with transmit commands. The test equipment shall not respond to any of the messages.

The pass criterion is a NR for each expected SMS.

The UUT shall then repeat the command messages and the test equipment shall respond starting at time T equal to 14.0 μ s, as shown in Fig. 7. The test shall be repeated with increasing values of T until the first NR occurs. Record time T when the first NR occurs.

5.2.1.5.2 No-Response Time-Out: RT-RT Transfers: The UUT shall send a minimum of nine RT-RT CMSs. The test equipment shall not respond to either command of three CMSs. The test equipment shall respond to the transmit command but not the receive command of three more CMSs, and finally, the test equipment shall not respond to the transmit command but shall respond to the receive command of the last three CMSs.

The pass criterion is a NR for each expected SMS.

The test shall be repeated with an increasing value of T for the transmitting response until the first NR occurs. Record time T when the first NR occurs. The test shall then be repeated with an increasing value of T for the receiving response until the first NR occurs. Record time T when the first NR occurs.

5.2.1.6 Error Injection: BC to RT Transfers: The purpose of these tests is to examine the UUT's response to specific errors in the message stream. Unless otherwise noted, the following test sequence shall be used for all error injection tests. The error to be encoded in Step 2 for a given message is specified in each test paragraph.

The pass criteria shall be as defined in each test paragraph. All commands and responses shall be recorded.

Test sequence:

Step 1: A CMS shall be sent by the UUT as specified in each paragraph.

Step 2: A SMS containing the specified error shall be sent to the UUT.

5.2.1.6.1 Parity: The purpose of these tests is to verify the UUT's capability of detecting parity errors embedded in different words within a message.

- 5.2.1.6.1.1 Status Word Validation for Transmit Commands: This test verifies the ability of the UUT to detect a parity error occurring in a status response to a transmit command word. The test sequence as defined in 5.2.1.6 shall be performed with a transmit command in Step 1 and a parity error encoded into the test equipment status word for test Step 2.

The pass criterion for this test is an ISMS for the transmitted SMS

- 5.2.1.6.1.2 Status Word Validation for Receive Commands: This test verifies the ability of the UUT to recognize a parity error occurring in a status response to a receive command word. The test sequence as defined in 5.2.1.6 shall be performed with a receive command in Step 1 and a parity error encoded in the test equipment status word for test Step 2.

The pass criterion for this test is an ISMS for the transmitted SMS

- 5.2.1.6.1.3 Data Word Validation: This test verifies the ability of the UUT to recognize a parity error occurring in a data word. The test sequence as defined in 5.2.1.6 shall be performed with a transmit command in Step 1 and a parity error encoded in a data word for Step 2. The SMS shall have the maximum number of data words that the UUT is designed to receive. The test sequence shall be sent N times, where N equals the number of data words sent. Individually, each data word shall have the parity bit inverted. Only one parity error is allowed per message.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.2 Word Length: These tests verify the ability of the UUT to recognize an error in word length occurring within a message. The test plan excludes testing of high bit count errors at the end of any message segment

- 5.2.1.6.2.1 Status Word Validation for Transmit Commands: This test verifies the ability of the UUT to recognize word length errors in a status response to transmit commands. The test sequence as defined in 5.2.1.6 shall be performed with a transmit command in Step 1 and with the status response word defined below for Step 2. Repeat this test with each of the status words defined below.

- a. Shorten the status word by one bit
- b. Shorten the status word by two bits
- c. Lengthen the status word by two bits
- d. Lengthen the status word by three bits

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.6.2.2 Status Word Validation for Receive Commands: This test verifies the ability of the UUT to recognize word length errors occurring in a status response to receive commands. The test sequence as defined in 5.2.1.6 shall be performed with a receive command in Step 1 and with the response word as defined below for Step 2. Repeat this test with each of the status words defined below.

- a. Shorten the status word by one bit
- b. Shorten the status word by two bits

The pass criterion for this test is an ISMS for each SMS transmitted.

5.2.1.6.2.3 Data Word Validation: This test verifies the ability of the UUT to recognize data word length errors. The test sequence as defined in 5.2.1.6 shall be performed with a transmit command in Step 1 and as defined below for Step 2. The response message shall be a status word with the maximum number of data words that the UUT is designed to receive. Repeat this test with each of the data words defined below.

- a. Shorten the data word by one bit
- b. Shorten the data word by two bits
- c. Lengthen the data word by two bits
- d. lengthen the data word by three bits

The test sequence of 5.2.1.6 shall be performed N times for A and B and NA times for C and D, where N equals the number of data words sent. High bit count errors shall not be tested in the last data word of the response. Only one data word shall be altered at a time. Steps A and B shall be performed for each data word in the message. Steps C and D shall be performed for each data word except the last.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.6.3 Bi-Phase Encoding: This test verifies the ability of the UUT to recognize bi-phase errors. A bi-phase encoding error is defined to be the lack of a zero crossing in the center of a bit time. A bi-phase error occurs as either a logic high or low for the duration of a bit time. Each bit location, except the sync period, of each word shall have a single W-phase error encoded into it. Only a single bi-phase error shall be injected for each message. Bi-phase errors shall be injected in such a way so as to not cause a parity error.

5.2.1.6.3.1 Status Word Validation for Transmit Commands: This test verifies the ability of the UUT to recognize bi-phase encoding errors in response to transmit commands. The test sequence as defined in 5.2.1.6 shall be performed with a transmit command in Step 1 and a bi-phase encoding error encoded into the status word for test Step 2. Only one bi-phase error is allowed per status word. A test set involves performing the test sequence 17 times, once for each bit location.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.3.2 Status Word Validation for Receive Commands: This test verifies the ability of the UUT to recognize bi-phase encoding errors in response to receive commands. The test sequence as defined in 5.2.1.6 shall be performed with a receive command in Step 1 and a bi-phase error encoded into a status word for test Step 2. Only one bi-phase error is allowed per command word. A test set involves performing the test sequence 17 times, once for each bit location.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.3.3 Data Word Validation: This test verifies the ability of the UUT to recognize bi-phase encoding errors in data words. The test sequence as defined in 5.2.1.6 shall be performed with a transmit command requesting the maximum number of data words that the UUT is designed to receive in Step 1 and a bi-phase error encoded into each data word in the response for Step 2. Individually, each bit location of each data word shall have a bi-phase error encoded into it. Only one bi-phase error is allowed for each message. A test set involves performing the sequence 17 times. The test set shall be repeated N times, where N equals the number of data words sent.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.4 Sync Encoding: This test verifies the ability of the UUT to recognize sync errors. The sync pattern, as defined for this test, is a waveform with six 0.5 ps divisions. The divisions are represented as a 1 or 0 to indicate the polarity of the divisions on the data bus. A proper status sync is represented as 111000 and a proper data sync is represented as 000111.

- 5.2.1.6.4.1 Status Word Sync Validation: This test verifies that the UUT rejects status words with invalid waveforms. The test sequence as defined in 5.2.1.6 shall be performed with a sync error encoded in a status word for Step 2. The following invalid sync patterns shall be used:

111100, 110000, 111001, 011000, 000111

The test sequence shall be performed for each of the above invalid sync patterns.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.4.2 Data Word Sync Validation: This test verifies that the UUT rejects invalid data sync waveforms. Perform the test sequence as defined in 5.2.1.6 with a transmit command requesting the maximum number of data words that the UUT is designed to receive in Step 1 and with a sync error encoded into each data word for Step 2. Only one data word per message shall have an invalid sync encoded into it. The following invalid sync patterns shall be used:

000011, 001111, 000110, 100111, 111000

The test sequence shall be performed N times for each of the invalid sync patterns, where N equals the maximum number of data words in the message. The test shall be performed so that each sync error shall occur with each data word.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.5 Response Message Length: These tests verify that the UUT properly detects an error when an incorrect number of data words is received.

- 5.2.1.6.5.1 Response to Receive Commands: This test verifies the ability of the UUT to detect an error condition if a data word is contiguous to a status word in response to a receive command. Perform the test sequence as defined in 5.2.1.6 with a receive command in Step 1 and a data word contiguously following a status word for Step 2.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.5.2 Response to Transmit Commands: This test shall verify that the UUT recognizes an error in the number of data words that are sent. The test sequence shall be performed as defined in 5.2.1.6 with a transmit command requesting the maximum number of data words in Step 1 and a data word count error in the SMS for Step 2. This response is a valid status word with different number of data words than specified in the command word. The test sequence shall be performed N times, where N equals the maximum number of data words. The first sequence shall have N+1 data words. The second sequence shall have N-1 data words and each of the remaining sequences shall remove one additional data word until the number of data words equals zero.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.6.6 Discontiguous Data: This test verifies that the UUT recognizes discontiguous data in a message. The test sequence shall be performed as defined in 5.2.1.6 with a transmit command in Step 1 with a 4.0 μ s data word gap error in a status response to a transmit command for Step 2. The gap is measured as in Fig. 7. This response shall be a status word with the maximum number of data words that the UUT is designed to receive with a gap between the status word and the first data word or between a data word pair. The test sequence shall be performed N times, where N equals the maximum number of data words. Each test sequence shall be performed with the gap appearing before a different data word. Only one gap time insertion is allowed per message.

The pass criterion is an ISMS for each SMS transmitted.

- 5.2.1.7 Error Injection: RT to RT Transfers: These tests verify the UUT's response to specific errors in an RT to RT message stream. Unless otherwise noted, the following test sequence shall be used for all RT-RT error injection tests. The no-response condition or encoding error for a given message is specified in each test paragraph.

The pass criteria are defined in each test paragraph.

All commands and responses shall be recorded.

Test sequence:

Step 1. A valid legal RT to RT command pair shall be sent by the UUT as specified in each paragraph.

Step 2. An RT response to the transmit command shall contain the specified error (if any) and shall be sent to the UUT by the test equipment.

Step 3. The specified RT response (if any) to the receive command shall be sent to the UUT by the test equipment.

Note that in the case of an RT to RT transfer, the SMS consists of an RT response to the transmit command followed by an RT response to the receive command.

- 5.2.1.7.1 Parity: These tests verify the ability of the UUT to detect parity errors embedded in different words within an RT to RT SMS.

- 5.2.1.7.1.1 Status Word Validation for Responses to RT to RT Transmit Commands: This test verifies the ability of the UUT to detect a parity error occurring in the status response to the transmit command in an RT to RT command. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with a parity error encoded in the test equipment status word response to the transmit command for Step 2 and a CS in Step 3.

The test sequence shall be performed the second time with a parity error encoded in the test equipment status word response to the transmit command for Step 2 and no response in Step 3.

In both cases, the pass criterion is an ISMS for the SMS transmitted.

- 5.2.1.7.1.2 Status Word Validation for Responses to RT to RT Receive Commands: This test verifies the ability of the UUT to recognize a parity error occurring in the status response to a receive command in an RT to RT command. The test sequence as defined in 5.2.1.7 shall be performed with no errors in Step 2, and a parity error in the test equipment status word response to the receive command for Step 3.

The pass criterion is an ISMS for the SMS transmitted.

- 5.2.1.7.1.3 RT to RT Data Word Validation: This test verifies the ability of the UUT to recognize a parity error occurring in a data word in an RT to RT SMS. The test sequence as defined in 5.2.1.7 shall be performed twice. The first sequence shall be performed with the RT to RT command pair requesting the maximum number of data words allowable by the UUT's design in Step 1, a parity error encoded in a data word for Step 2, and a CS in Step 3.

The test sequence shall be performed the second time with a parity error in a data word for Step 2 and no response in Step 3.

Each test sequence of 5.2.1.7 shall be sent N times, where N equals the number of data words sent. Only one data word shall have the parity bit inverted for each message sent, with a different data word being altered for each execution of the test sequence.

The pass criterion is an ISMS for the SMS transmitted.

- 5.2.1.7.2 Word Length: These tests verify the ability of the UUT to recognize word length errors within any word in an RT to RT SMS. The test plan excludes testing of high bit count errors at the end of any word sequence.

5.2.1.7.2.1 Status Word Validation for Responses to RT to RT Transmit Commands: This test verifies the ability of the UUT to recognize word length errors occurring in the status response to an RT to RT transmit command. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with the status response word defined below for Step 2 and a CS for Step 3. The test sequence shall be performed a second time with a no-response in Step 3. Repeat those sequences with each of the status words defined below.

- a. Shorten status word by one bit.
- b. Shorten status word by two bits.
- c. Lengthen status word by two bits.
- d. Lengthen status word by three bits.

The pass criterion for this test is an ISMS for each SMS transmitted.

5.2.1.7.2.2 Status Word Validation for Responses to RT to RT Receive Commands: This test verifies the ability of the UUT to recognize word length errors occurring in the status response to an RT to RT receive command. The test sequence as defined in 5.2.1.7 shall be performed with no errors in Step 2 and with the status word defined below used for Step 3. Repeat this test with each of the status words defined below.

- a. Shorten status word by one bit.
- b. Shorten status word by two bits.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.7.2.3 Data Word Validation: This test verifies the ability of the UUT to recognize data word length errors. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with the RT to RT command pair requesting the maximum number of data words allowable by the UUT's design in Step 1, a data word from the list below in Step 2, and a CS in Step 3.

The test sequence shall be performed a second time with no response in Step 3. Repeat these sequences using each data word shown below.

- a. Shorten data word by one bit. 
- b. Shorten data word by two bits.
- c. Lengthen data word by two bits.
- d. Lengthen data word by three bits.

Each test sequence of 5.2.1.7 shall be performed N times for A and B and N-1 times for C and D, where N equals the number of data words sent. High bit count errors shall not be tested in the last data word of the response. Only one data word shall be altered at a time. Steps A and B shall be performed for each data word in the message. Steps C and D shall be performed for each data word except the last.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.7.3 Bi-Phase Encoding: These tests verify the ability of the UUT to recognize bi-phase errors within any word in an RT to RT SMS. A bi-phase error is defined to be the lack of a zero crossing in the center of a bit time. A bi-phase error occurs as either a logic high or low for the duration of a bit time. Each bit location, except the sync period, of each word shall have a single bi-phase error encoded into it. Only a single bi-phase error shall be injected for each message. Bi-phase errors shall be injected in such a way as to not cause a parity error.

5.2.1.7.3.1 Status Word Validation for Responses to RT to RT Transmit Commands: This test verifies the ability of the UUT to detect bi-phase encoding errors occurring in the status response to an RT to RT transmit command. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with a bi-phase error encoded in the status word for test Step 2 and a CS in test Step 3.

The test sequence shall be performed the second time with a no-response in Step 3. Only one bi-phase error is allowed per status word. A test set involves performing each test sequence 17 times, once for each bit location.

The pass criterion is an ISMS for the SMS transmitted.

5.2.1.7.3.2 Status Word Validation for Responses to RT to RT Receive Commands: This test verifies the ability of the UUT to recognize bi-phase encoding errors occurring in the status response to an RT to RT receive command. The test sequence as defined in 5.2.1.7 shall be performed with no errors in Step 2 and a bi-phase error in the status word for Step 3. only one bi-phase error is allowed per status word. A test set involves performing the test sequence 17 times, once for each bit location.

The pass criterion is an ISMS for the SMS transmitted.

5.2.1.7.3.3 Data Word Validation: This test verifies the ability of the UUT to recognize bi-phase encoding errors occurring in a data word in an RT to RT SMS. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with the RT to RT command pair requesting the maximum number of data words allowable by the UUT's design in Step 1, a bi-phase error encoded in a data word for Step 2, and a CS in Step 3.

The test sequence shall be performed the second time with a no-response in Step 3.

Each test sequence of 5.2.1.7 must be sent N times, where N equals the number of data words sent. Only one data word shall have a bi-phase encoding error for each message sent, with a different data word being altered for each execution of the test sequence.

The pass criterion is an ISMS for the SMS transmitted.

5.2.1.7.4 Sync Encoding: This test verifies the ability of the UUT to recognize sync errors in the status and data word syncs in an RT to RT SMS. The sync pattern, as defined for this test, is a waveform with six 0.5 μ s divisions. Each division is represented as a 1 or 0 to indicate the polarity of the divisions on the data bus. A proper status sync is represented as 111000 and a proper data sync is represented as 000111.

5.2.1.7.4.1 Status Word Sync Validation for Response to RT to RT Transmit Commands: This test verifies the ability of the UUT to recognize sync waveforms in the status response to an RT to RT transmit command. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with a sync error (as shown below) encoded in the status word for Step 2, and a CS in Step 3. The second test sequence shall be performed with a no response in Step 3. The following invalid sync patterns shall be used:

111100, 110000, 111001, 011000, 000111

Each test sequence shall be performed for each of the above invalid sync patterns.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.7.4.2 Status Word Sync Validation for Response to RT to RT Receive Commands: This test verifies the ability of the UUT to recognize invalid sync waveforms in the status response to an RT to RT receive command. The test sequence as defined in 5.2.1.7 shall be performed with no errors in Step 2, and a sync error (as shown below) encoded in the status word for Step 3. The following invalid sync patterns shall be used:

111100, 110000, 111001, 011000, 000111

The test sequence shall be performed for each of the above invalid sync patterns.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.7.4.3 Data Word Sync Validation: This test verifies the ability of the UUT to recognize invalid data sync waveforms in an RT to RT SMS. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with the RT to RT command pair requesting the maximum number of data words allowable by the UUT's design. A sync error shall be encoded into each data word for Step 2, and a CS in Step 3. The test sequence shall be performed the second time with a no response in Step 3. Only one data word per SMS shall have an invalid sync encoded into it. The following invalid sync patterns shall be used:

000011, 001111, 000110, 100111, 111000

5.2.1.7.4.3 (Continued):

Each test sequence shall be performed N times for each of the above invalid sync patterns, where N equals the maximum number of data words in the message.

The test shall be performed so that each sync error shall occur with each data word.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.7.5 Response Message Length: This test verifies the ability of the UUT to recognize message length errors when an incorrect number of data words are transmitted by either RT in an RT to RT SMS.

5.2.1.7.5.1 Message Length Validation for RT to RT Transmit Commands: This test verifies the ability of the UUT to detect an error condition when an incorrect number of data words is transmitted in response to an RT to RT transmit command. The test sequence as defined in 5.2.1.7 shall be performed twice. The first test sequence shall be performed with the RT to RT command pair requesting the maximum number of data words allowable by the UUT's design. A data word count error shall be injected into the SMS for Step 2, and a CS in Step 3.

The test sequence shall be performed the second time with a no response in Step 3.

Each test sequence shall be performed N times, where N equals the maximum number of data words. The first sequence shall have N+1 data words, the second sequence shall have N-1 data words and each of the remaining sequences shall remove one additional data word until the number of data words equals zero.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.7.5.2 Message Length Validation for RT to RT Receive Commands: This test verifies the ability of the UUT to detect an error condition if a data word is contiguous to a status word in response to the RT to RT receive command. The test sequence as defined in 5.2.1.7 shall be performed with no errors in Step 2 and one data word contiguously following the status word for Step 3.

The pass criterion is an ISMS for each SMS transmitted.

5.2.1.7.6 Discontiguous Data: This test verifies that the UUT recognizes discontiguous data in an RT to RT SMS. The test sequence as defined in 5.2.1.7 shall be performed with the transmit command in the RT to RT command pair requesting the maximum number of data words allowable by the UUT's design. A 4.0 μ s gap shall be injected either between the status word and the first data word or between a data word pair for Step 2 and no errors in Step 3. The gap is measured as in Fig. 7.

The test sequence shall be performed N times, where N equals the maximum number of data words. Each test sequence shall be performed with the gap appearing before a different data word. Only one gap time insertion is allowed per message.

The pass criterion is an ISMS for each SMS transmitted.

5.2.2 Optional Operation: This section provides for testing the optional requirements of MIL-STD-1553. If a UUT implements any of the options, it shall be tested in accordance with the test herein identified for the option.

5.2.2.1 Mode Commands: This section verifies that the UUT properly implements the processing of mode commands. All optional mode commands defined in MIL-STD-1553 shall be tested both with normal responses from the test equipment and with selected error responses.

5.2.2.1.1 Dynamic Bus Control: This test verifies the ability of the UUT to initiate a dynamic bus control mode code sequence, identify the RT response, and take appropriate action, such as:

1. Cause the UUT to send an MCO. The test equipment shall respond with a valid status word with the DBACC bit set to zero. The pass criteria are a VSMS and verification that the UUT detects that the DBACC bit was not set.
2. Cause the UUT to send an MCO. The test equipment shall respond with a valid status word with the DBACC bit set to one. The pass criteria are a VSMS and verification that the UUT detects that the DBACC bit was set.
3. Cause the UUT to send an MCO. The test equipment shall not respond to the command. The pass criterion is an NR.
4. Cause the UUT to send an MCO. The test equipment shall respond with a status word with the DBACC bit set and invalid parity. The pass criterion is an ISMS.
5. Cause the UUT to send an MCO. The test equipment shall respond with a valid status word with the wrong RT address. The pass criterion is an ISMS.

5.2.2.1.1 (Continued):

6. Cause the UUT to send an MCO. The test equipment shall respond with a valid, legal status word with the DBACC bit set and a contiguous valid data word.
The pass criterion is an ISMS.
7. Cause the UUT to send an MCO. The test equipment shall respond with a legal status word with the DBACC bit set on an alternate bus.
The pass criterion is an NR or an ISMS.

5.2.2.1.2 Mode Commands Without Data Words: In the following tests, each mode command from MCI through MC8, as defined in Table 1, shall be tested as follows:

1. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid, legal status word.
The pass criterion is a VSMS for the SMS.
2. Cause the UUT to send the selected mode command. The test equipment shall not respond to the command.
The pass criterion is an NR.
3. Cause the UUT to send the selected mode command. The test equipment shall respond with a legal status word with invalid parity.
The pass criterion is an ISMS.
4. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid status word with the wrong RT address.
The pass criterion is an ISMS.
5. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid status word with a contiguous valid data word.
The pass criterion is an ISMS.
6. Cause the UUT to send the selected mode command. The test equipment shall respond with a status word on an alternate bus.
The pass criterion is an NR or an ISMS.

5.2.2.1.3 Transmit Mod Commands With Data Words: In the following test, each of the mode commands, MC16, MC18, and MC19, as shown in Table 1, shall be tested as follows:

1. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid status word and one contiguous data word.
The pass criterion is a VSMS.
2. Cause the UUT to send the selected mode command. The test equipment shall not respond to the command.
The pass criterion is an NR.
3. Cause the UUT to send the selected mode command. The test equipment shall respond with a status word with invalid parity and a valid data word.
The pass criterion is an ISMS.
4. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid status word with the wrong RT address followed by a valid data word.
The pass criterion is an ISMS.
5. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid status word without a data word.
The pass criterion is an ISMS.
6. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid status word with two contiguous, valid data words.
The pass criterion is an ISMS.
7. Cause the UUT to send the selected mode command. The test equipment shall respond with a valid status word and a legal data word with a parity error.
The pass criterion is an ISMS.
8. Cause the UUT to send the selected mode command. The test equipment shall respond on an alternate bus with a status word and one contiguous valid data word.
The pass criterion is an NR or an ISMS.

~~~~~a~~~~~a~~~~~a~~~~~a~~~~~a~~~~~

5.2.2.1.4 Receive Mode Commands With Data Words: In the following tests, each of the mode commands, MC17, MC20, and MC21, as shown in Table 1, shall be tested as follows:

1. Cause the UUT to send the selected mode command and contiguous data word. The test equipment shall respond with a valid status word.  
The pass criterion is a VSMS.
2. Cause the UUT to send the selected mode command and data word. The test equipment shall not respond to the command.  
The pass criterion is an NR.
3. Cause the UUT to send the selected mode command and data word. The test equipment shall respond with a legal status word with invalid parity.  
The pass criterion is an ISMS.
4. Cause the UUT to send the selected mode command and data word. The test equipment shall respond with a valid status word with the wrong RT address.  
The pass criterion is an ISMS.
5. Cause the UUT to send the selected mode command and data word. The test equipment shall respond with a valid status word with a contiguous, valid data word.  
The pass criterion is an ISMS.
6. Cause the UUT to send the selected mode command and data word. The test equipment shall respond with a status word on an alternate bus.  
The pass criterion is an NR or an ISMS.

5.2.2.2 Status Word: The following tests are intended to confirm that the UUT is capable of detecting all of the bits of the status word for each message format. The bits shall be tested one at a time by successively setting each of the eleven status code bits (bits 9-19) to a logic one with the remaining status code bits set to a logic zero. In each case, verify the ability of the UUT to detect each bit and verify any special function associated with the bit that the UUT specification requirement supports in accordance with the requirements of MIL-STD-1553. In the event of a failure, record the corresponding status word bit position.

5.2.2.2.1 Status Bits. Receive Commands: Cause the UUT to send a sequence of eleven receive commands with the appropriate number of data words. Cause the test equipment to respond to each of the commands in the sequence with a valid status word with a different bit set in each status code field.

The pass criterion is the verification of bit detection.

5.2.2.2.2 Status Bits, Transmit Commands: Cause the UUT to send a sequence of eleven transmit commands. Cause the test equipment to respond to each of the commands in the sequence with a valid status word, and the appropriate number of data words (no data words for the "busy bit set" case). The status words shall have a different bit set in each status code field. The pass criterion is the verification of bit detection.

5.2.2.2.3 Status Bits, RT to RT Commands, Transmitter Status: Cause the UUT to send a sequence of eleven RT to RT command pairs. Cause the test equipment to respond to each of the transmit commands in the sequence with a valid status word and the appropriate number of data words (no data words for the "busy bit set" case). The status word for the transmitter response shall have a different bit set in each status code field. The test equipment shall also respond to the receive command with a valid status word with all eleven status code bits set to a logic zero.

The pass criterion is the verification of bit detection.

5.2.2.2.4 Status Bits, RT to RT Commands, Receiver Status: Cause the UUT to send a sequence of eleven RT to RT command pairs. Cause the test equipment to respond to each of the transmit commands in the sequence with a valid status word with all eleven status code bits set to a logic zero, and the appropriate number of data words. The test equipment shall also respond to each of the receive commands in the sequence with a status word with a different bit set in each status code field.

The pass criterion is the verification of bit detection.

5.2.2.2.5 Status Bits, Mode Command: For this test, cause the UUT to send a sequence of eleven mode commands. Cause the test equipment to respond to each of the mode commands in the sequence with a valid status word and valid data word, if required. The status words shall have a different bit set in each status code field.

The pass criterion is the verification of bit detection. Repeat this test for each mode command supported by the UUT.

5.2.2.3 Broadcast Command: This test verifies that the UUT does not expect a response to a broadcast command, a command with the RT address field set to all ones, or that the UUT's operation is not disrupted if a status response occurs. Each type of broadcast command shall be tested.

5.2.2.3.1 Broadcast Receive Commands: Cause the UUT to send a receive broadcast command. The test equipment shall not respond to this command.

The pass criterion is verification that the UUT registered an NR as the correct response.

- 5.2.2.3.2 Broadcast Receive Command With Response: Cause the UUT to send a receive broadcast command. Cause the test equipment to respond to this command with a valid status word.

The pass criterion is an ISMS detected by the UUT.

- 5.2.2.3.3 Broadcast RT-RT Command: This test verifies that the UUT does not expect a response from receiving RT in broadcast RT-RT transfer.

- 5.2.2.3.3.1 Broadcast RT-RT Command With No-Response: Cause the UUT to send a broadcast RT-RT command. The test equipment shall not respond.

The pass criterion is the verification that the UUT registered an NR.

- 5.2.2.3.3.2 Broadcast RT-RT Command With Response: Cause the UUT to send a broadcast RT-RT command. Cause the test equipment to respond with a status word and the appropriate number of data words representing the transmitting RT's response. Cause the test equipment to follow this with a status word representing a response from a receiving RT.

The pass criterion is an ISMS detected by the UUT.

- 5.2.2.3.3.3 Normal Broadcast RT-RT Command: Cause the UUT to send a broadcast RT-RT command. Cause the test equipment to respond with a status word and the appropriate number of data words representing the transmitting RT's response.

The pass criterion is a VSMS detected by the UUT.

- 5.2.2.3.4 Broadcast Mode Commands: Perform this test for the following broadcast mode commands that can be sent by the UUT: MC1, MC3, MC4, MC5, MC6, MC7, MC8, MC17, MC20, MC21. Let the UUT send each broadcast mode command. The test equipment shall not respond to these commands.

The pass criterion is verification that the UUT registered an NR as the correct response.

- 5.2.2.3.5 Broadcast Mode Command With Response: Cause the UUT to send each of the following legal broadcast mode commands that are supported by the UUT design specification: MC1, MC3, MC4, MC5, MC6, MC7, MC8, MC17, MC20, MC21. Cause the test equipment to respond to each command with a valid status response. The pass criteria shall be an ISMS detected by the UUT for each command.

5.2.2.4 Response On Alternate Bus: This test verifies that the UUT detects a response to a command on the opposite bus from which the command was issued as an erroneous response. During this test, the UUT shall send out a command and the test equipment shall respond to the command on each alternate bus. This test shall be conducted with a transmit command, a receive command, and an RT -RT command pair. The RT-RT case shall be conducted for each bus, first for the transmitter response on each alternate bus, then for the receiver response on each alternate bus. The UUT shall not accept a response on an alternate bus.

The pass criterion for each case is an NR or ISMS for each SMS on an alternate bus.

5.3 Noise Rejection: This test verifies the UUT's ability to operate in the presence of noise. Fig. 8 depicts a typical configuration for conducting the noise rejection test. The maximum word error rate for a BC is one part in  $10E7$ . While performing this test, all words received by the UUT shall be in the presence of an additive white Gaussian noise distributed over a bandwidth of 1.0 KHz to 4.0 MHz at an RMS amplitude of 140 mV for transformer coupled stubs or 200 mV for direct coupled stubs measured at point A of Fig. 8. This test shall be conducted with a signal level of 2.1 V peak-to-peak, line-to-line, for direct coupled stubs measured at point A of Fig. 8. For the purposes of this test, an error shall include any fault that causes the UUT to detect an invalid word. The noise test shall run continuously until the total number of all words received by the UUT exceeds the required number for acceptance of the UUT, or is less than the required number for rejection of the UUT, as specified in Table 2. All data words used in the tests shall contain random bit patterns. These bit patterns shall be unique for each data word in a message and shall change randomly from message to message.

The pass/fail criteria are contained in Table 2.

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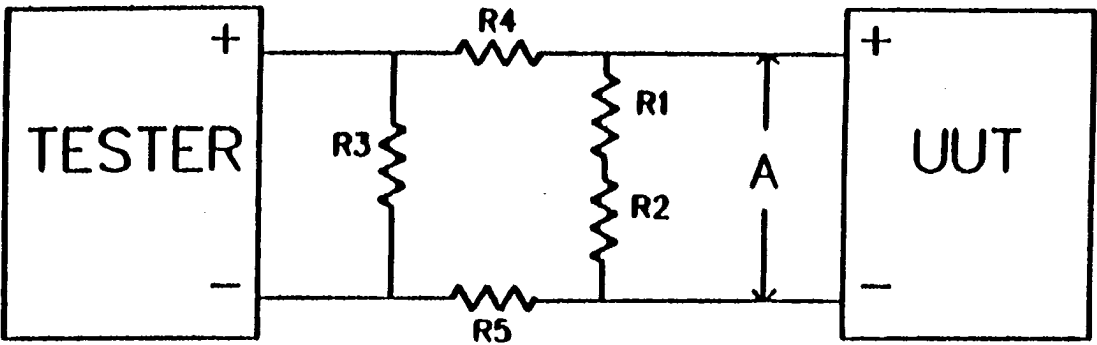
PREPARED BY SUBCOMMITTEE AS-1A, AVIONICS SYSTEMS,
OF SAE COMMITTEE AS-1, AVIONICS/ARMAMENT INTEGRATION

TABLE 1 - Mode Codes

MC#	MODE CODE FIELD	FUNCTION
MC0	00000	Dynamic bus control
MC1	00001	Synchronize
MC2	00010	Transmit status word
MC3	00011	Initiate self-test
MC4	00100	Transmitter shutdown
MC5	00101	Override transmitter shutdown
MC6	00110	Inhibit terminal flag bit
MC7	00111	Override inhibit terminal flag bit
MC8	01000	Reset remote terminal
MC16	10000	Transmit vector word
MC17	10001	Synchronize (with data word)
MC18	10010	Transmit last command
MC19	10011	Transmit BIT word
MC20	10100	Selected transmitter shutdown
MC21	10101	Override selected transmitter shutdown

TABLE 2 -Criteria for Acceptance or Rejection of a Bus
Controller for the Noise Rejection Test Total Number
of Words Received by the Terminal
(in multiples of 10E7)

NO OF ERRORS	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	4.40
1	N/A	5.21
2	N/A	6.02
3	N/A	6.83
4	N/A	7.64
5	N/A	8.45
6	0.45	9.27
7	1.26	10.08
8	2.07	10.89
9	2.88	11.70
10	3.69	12.51
11	4.50	13.32
12	5.31	14.13
13	6.12	14.94
14	6.93	15.75
15	7.74	16.56
16	8.55	17.37
17	9.37	18.19
18	10.18	19.00
19	10.99	19.81
20	11.80	20.62
21	12.61	21.43
22	13.42	22.24
23	14.23	23.05
24	15.04	23.86
25	15.85	24.67
26	16.66	25.48
27	17.47	26.29
28	18.29	27.11
29	19.10	27.92
30	19.90	28.73
31	20.72	29.54
32	21.53	30.35
33	22.34	31.16
34	23.15	31.97
35	23.96	32.78
36	24.77	33.00
37	25.58	33.00
38	26.39	33.00
39	27.21	33.00
40	28.02	33.00
41	33.00	N/A



	DIRECT COUPLED	TRANSFORMER COUPLED
R1,R2	35Ω +2%	70Ω +2%
R3, R4, R5	20	46.5
	100	93.1

FIGURE 1 - General Test Configuration

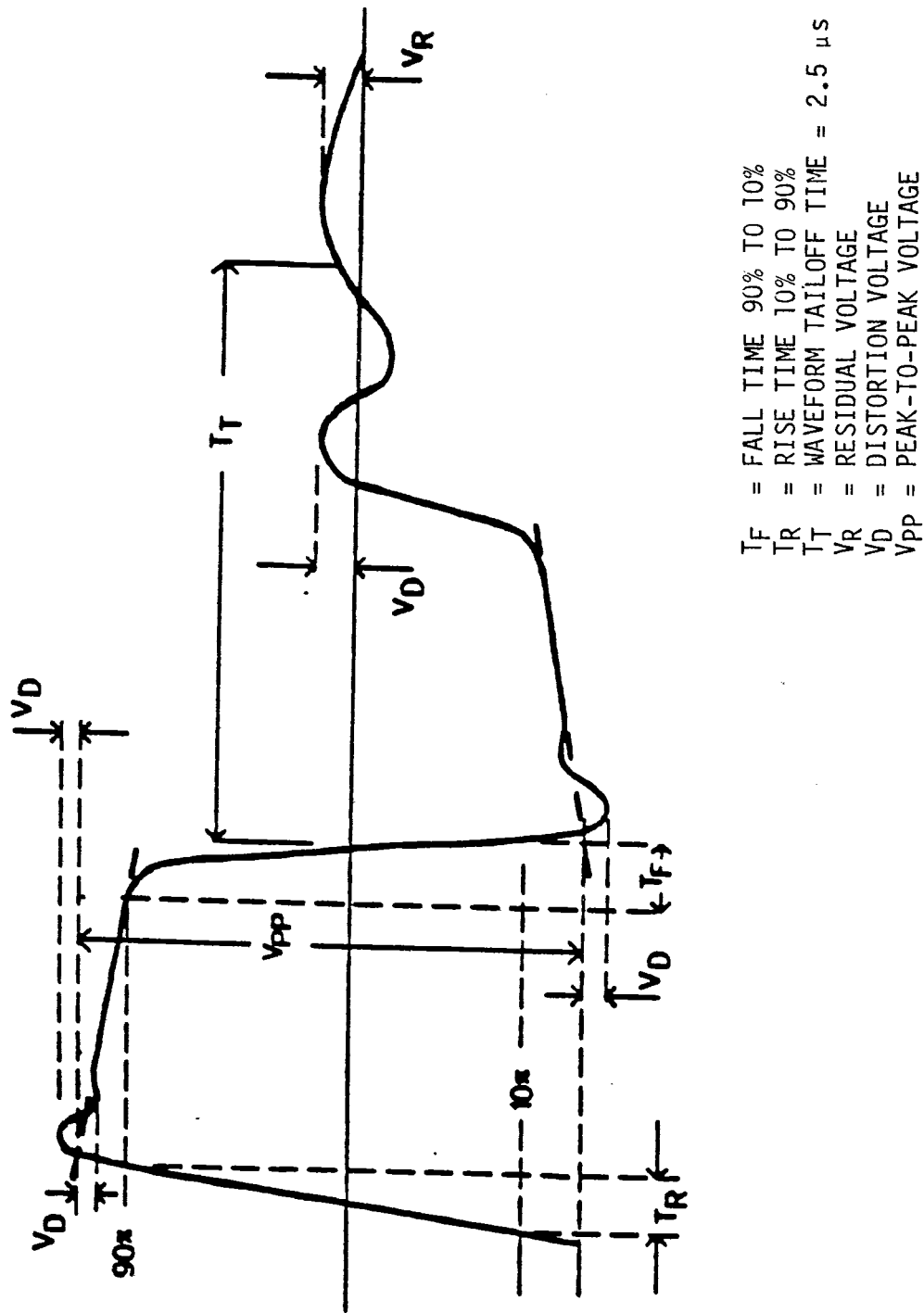


FIGURE 2 - Waveform Measurements

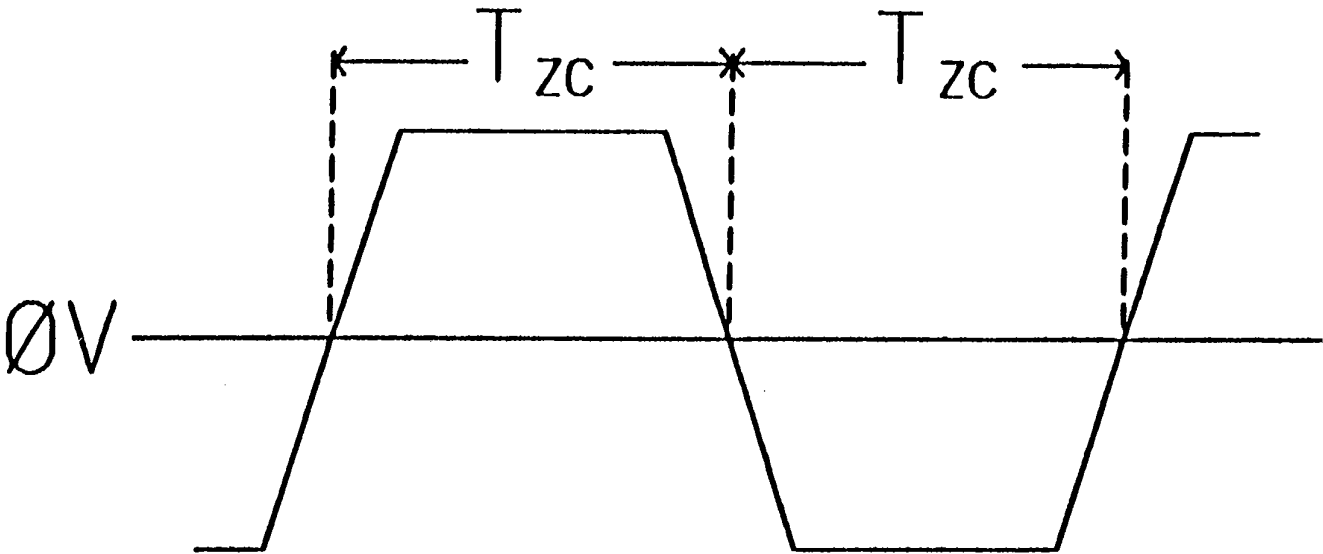


FIGURE 3 - Zero Crossing Interval Measurements

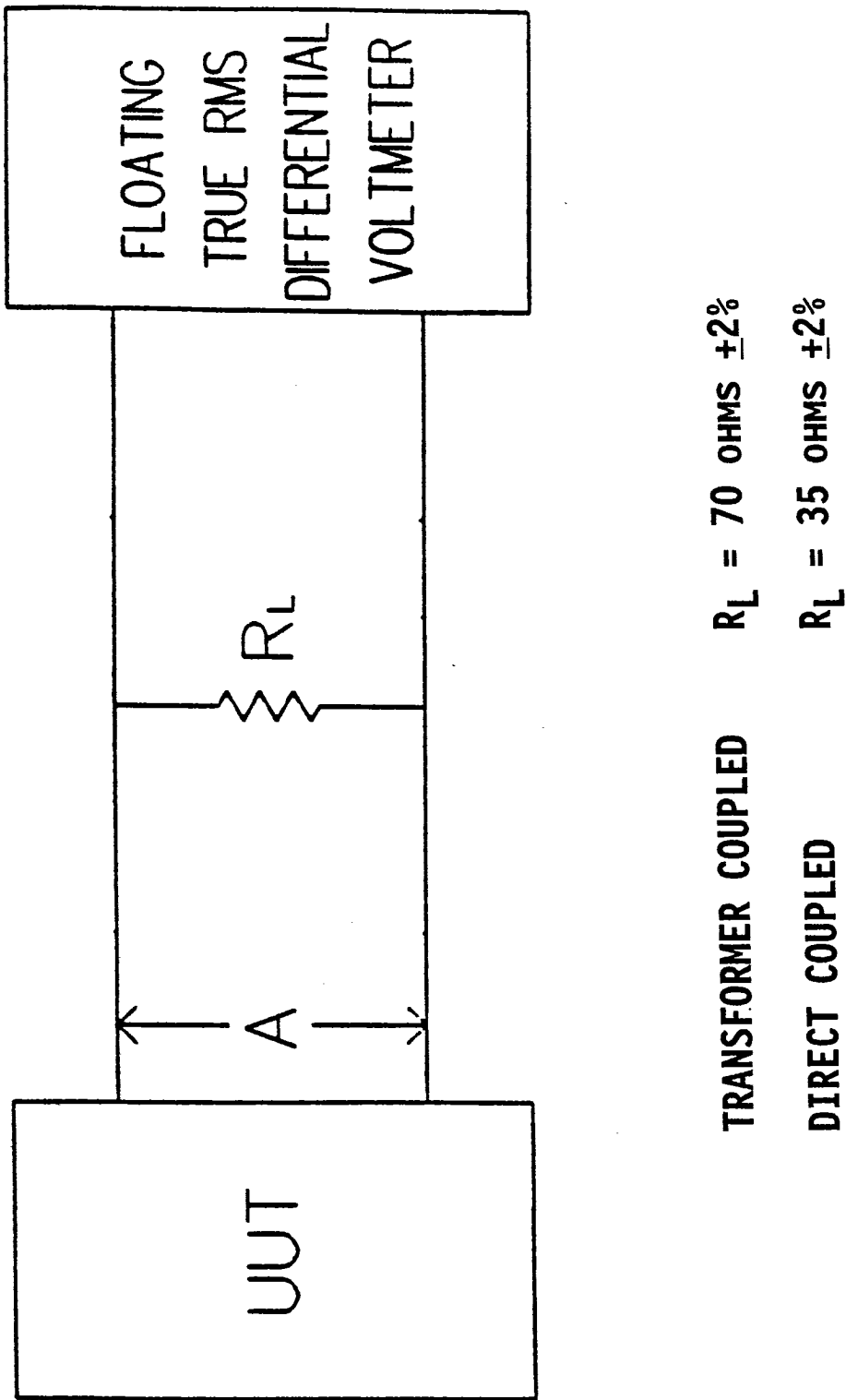


FIGURE 4 -- Output Noise Configuration

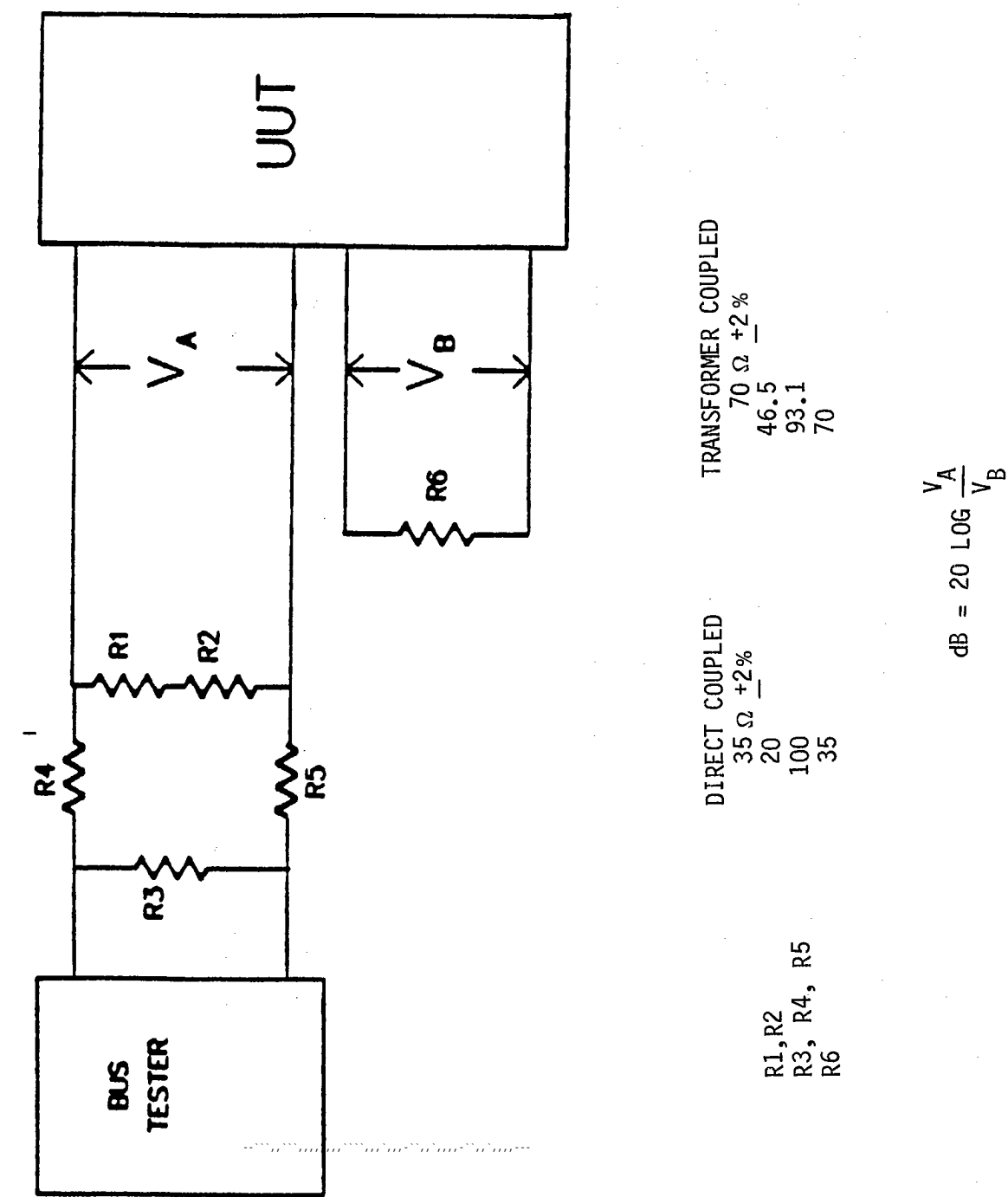
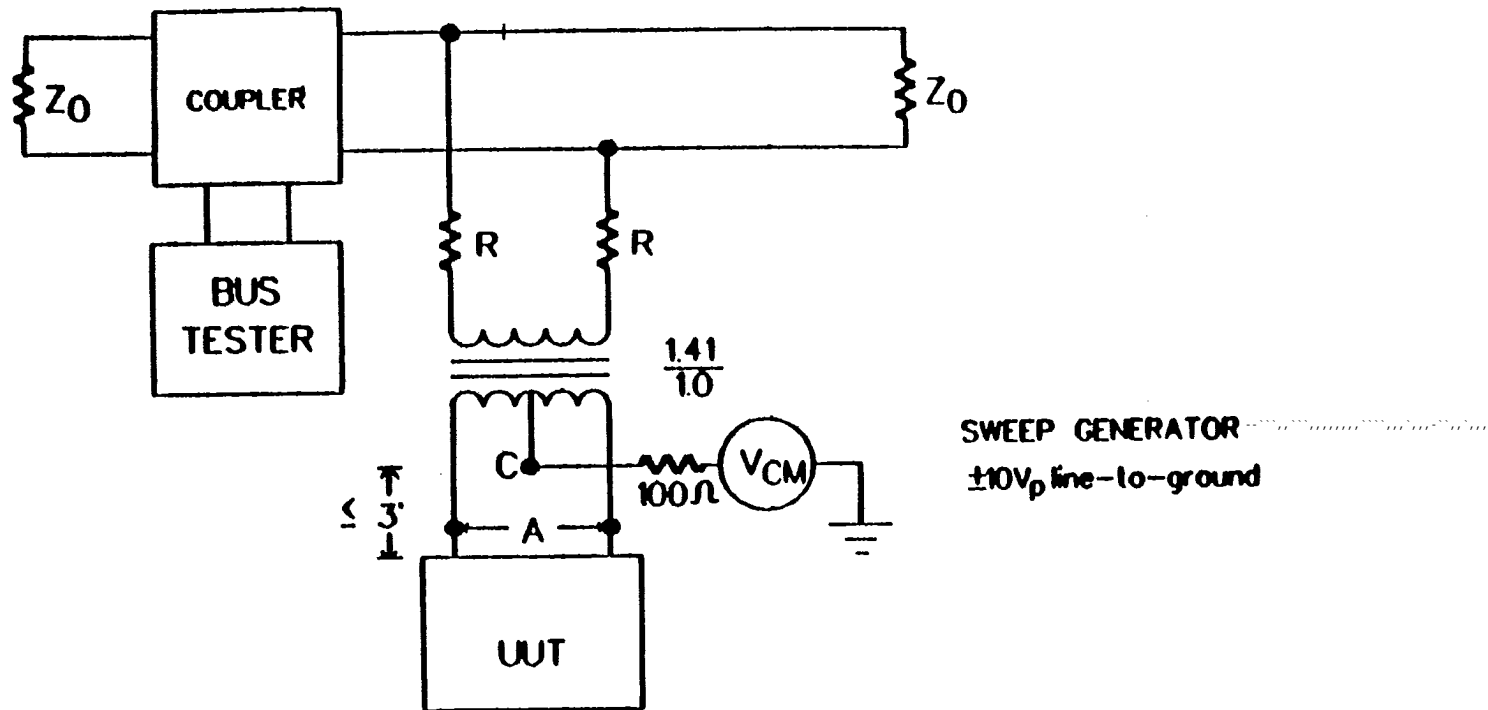
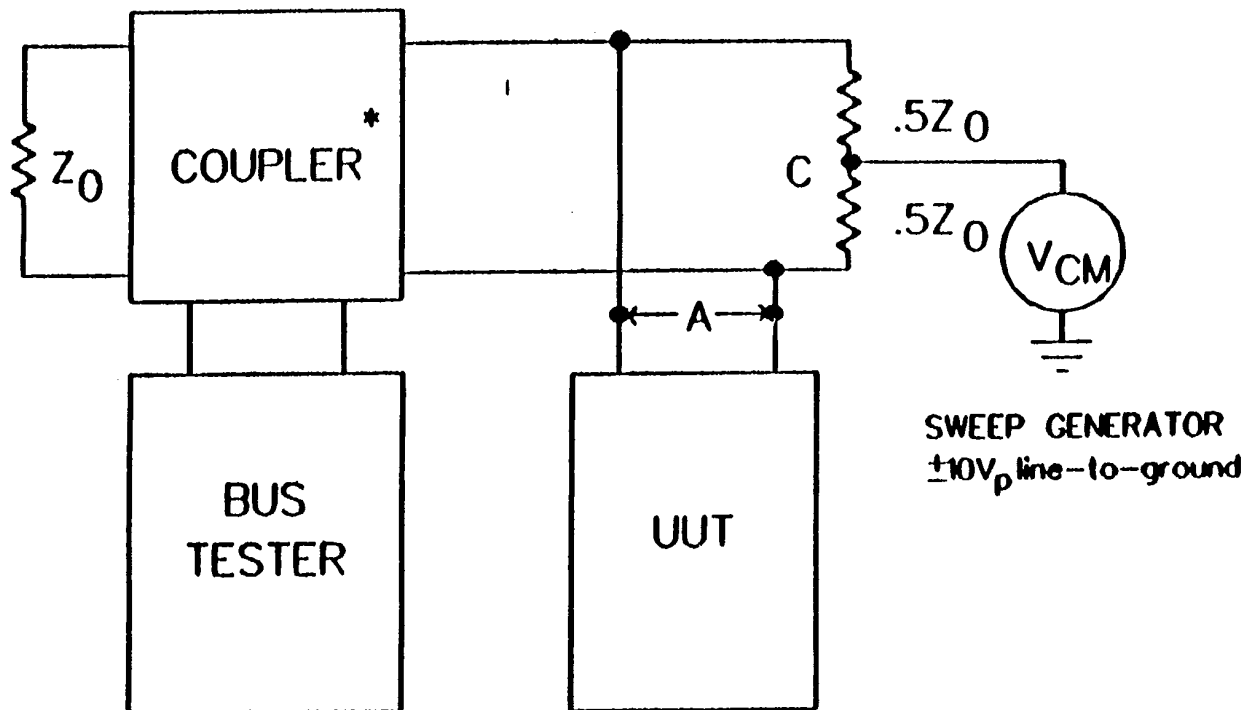


FIGURE 5 - Output Isolation



$R = 0.75 Z_0 \pm 2\%$
 Z_0 - Selected Cable Nominal Characteristic Impedance

FIGURE 6A - Transformer Coupled Common Mode Configuration



* - Transformer Coupled Stub Must Be Used
 Z_0 - Selected Cable Nominal Characteristic Impedance

FIGURE 6B - Direct Coupled Common Mode Configuration

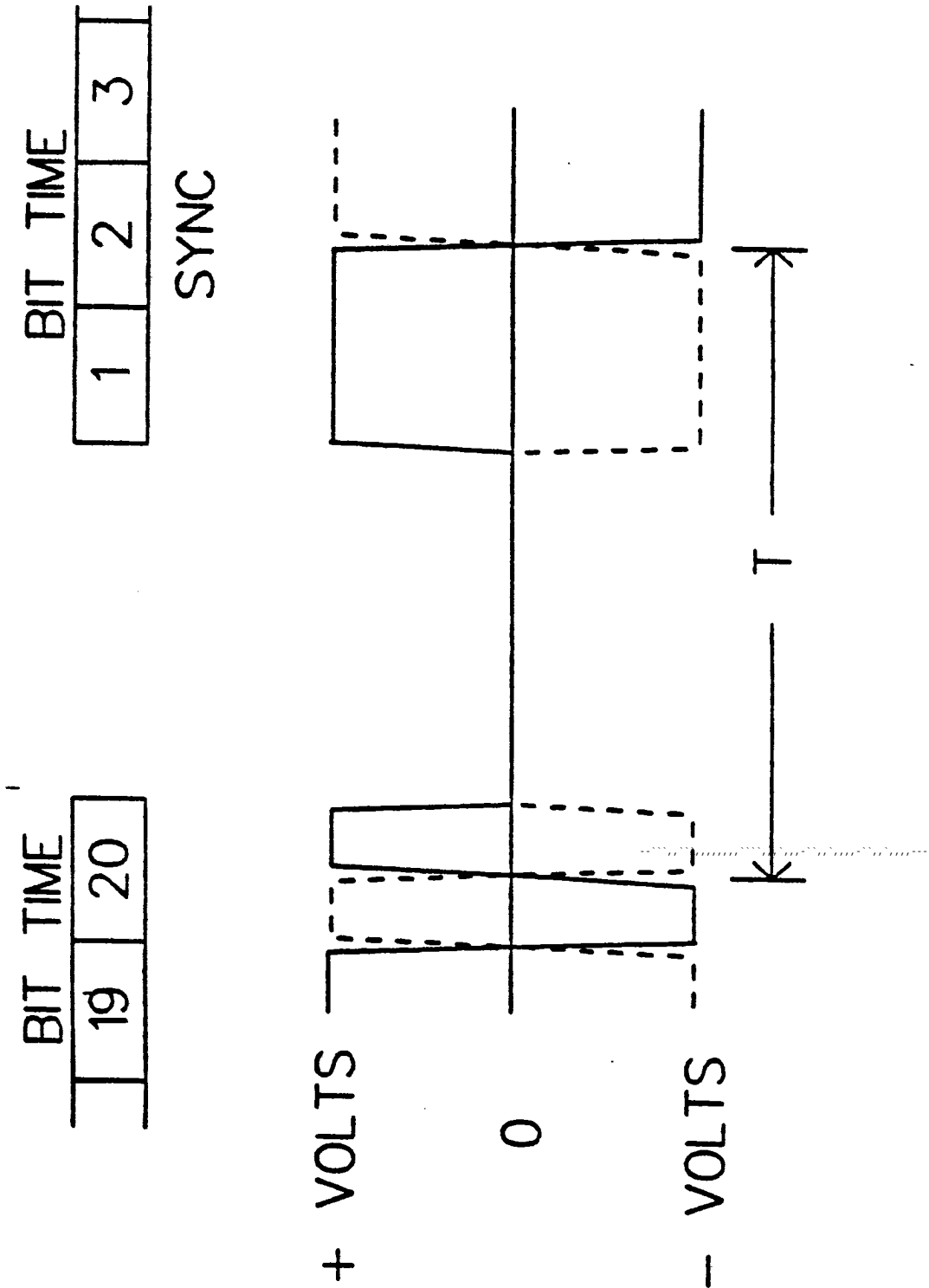


FIGURE 7 – Gap Time Measurement

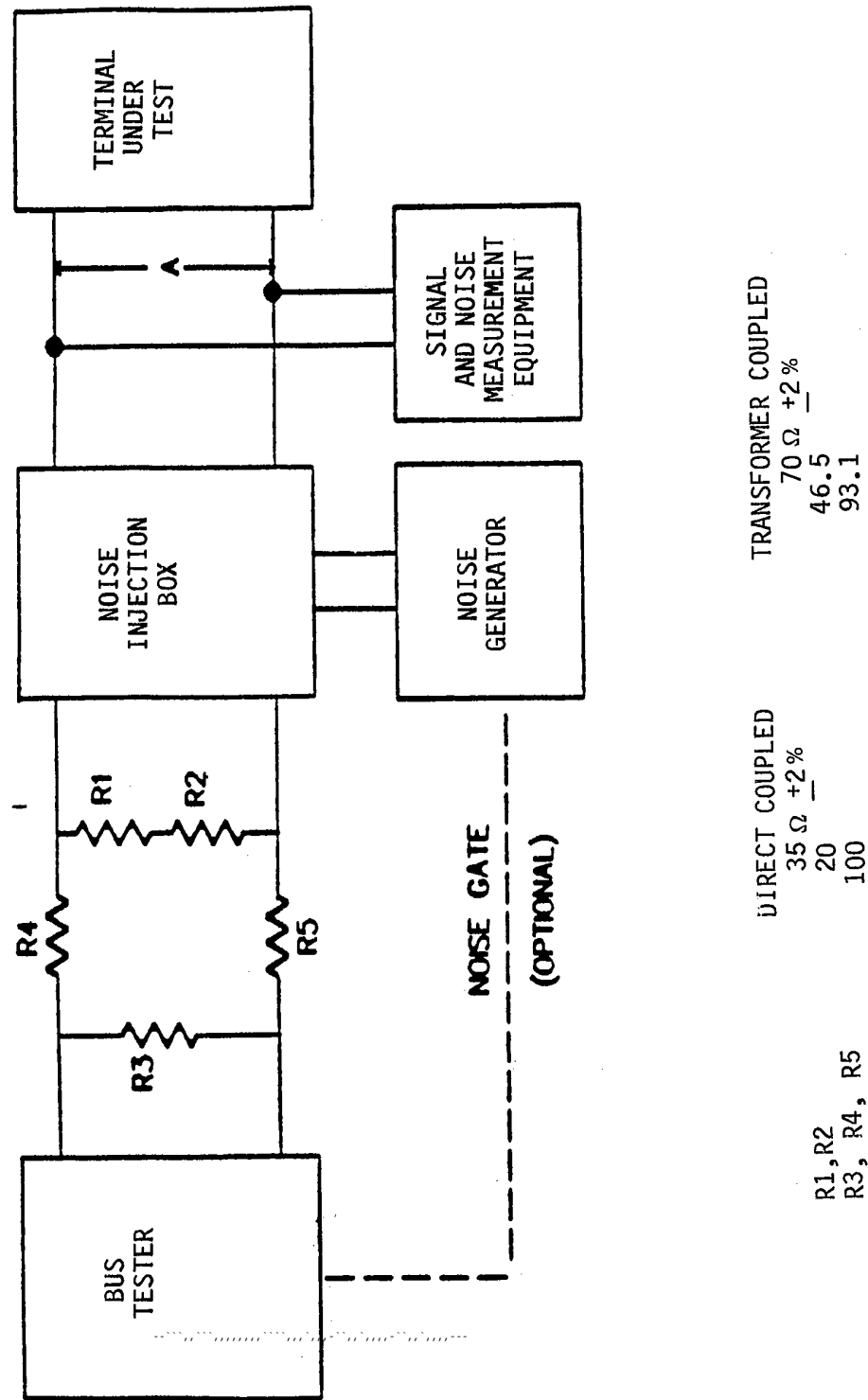


FIGURE 8 - Configuration for Noise Rejection Test

APPENDIX A
TEST PLAN CHANGES FOR MIL-STD-1553B, NOTICE 2 BCs

For bus controllers designed to comply with MIL-STD-1553B, Notice 2, the following changes shall be made for tests or pass criteria, or both, in this document:

A1. Connector Polarity: Add 4.2 as follows:

4.2 Connector Polarity: When performing tests for UUTs that use concentric connectors or inserts for each bus, observe polarity conventions in connecting to the UUTs such that the center pin of the connector or insert shall be used for the high (positive) Manchester bi-phase signal and the inner ring shall be used for the low (negative) Manchester bi-phase signal.

A2. Mode Codes: The tests in 5.2.2.1 and subparagraphs are mandatory. Test delineated in the subparagraphs shall be performed twice, once with the subaddress/mode code of the command word set to 00000 and once set to 11111.

A3. Broadcast: The tests in 5.2.2.3.1, 5.2.2.3.2 and 5.2.2.3.3 are mandatory.

The tests in 5.2.2.3.4 for mode commands MC4, MCS and MC8 are mandatory; for mode commands MC1, MC3, MC6, MC7, MC20 and MC21, the tests are optional.

The tests in 5.2.2.3.5 for mode commands MC4, MCS and MC8 are mandatory; for mode commands MCI, MC3, MC6, MC7, MC20 and MC21, the tests are optional.